



PROGRAMA CENTELHA RJ · FAPERJ · 2026

A voz da IA, com sotaque brasileiro.

Uma voz de IA que conversa, tem emoção e fala com o sotaque e o jeito do Brasil. A começar pelo carioca.



QUEM ESTÁ CONSTRUINDO

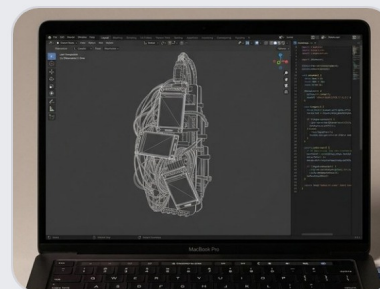
Pedro Cormann.



ENG. DE MACHINE LEARNING · TECH ARTIST

Construindo o TTS-ptbr sozinho: modelo, dados, treino e produto. 5+ anos de IA, visão computacional e sensores para TikTok, Meta, Coca-Cola, Samsung, L'Oréal, Nubank e Netflix. Prototipando guia de áudio com óculos Ray-Ban Meta para museus. NVIDIA Inception · LNCC · Eng. PUC-Rio.

DE VR E ÓCULOS INTELIGENTES A HARDWARE E DRONES



FORMAÇÃO EM ML · TENSORFLOW + TF DEPLOYMENT (COURSERA) · DATA SCIENTIST/ML + AI ENGINEER (CODECADEMY) · DEEP LEARNING (NVIDIA)

Domino o projeto de ponta a ponta, com autonomia técnica total.

A IA já fala português brasileiro. Mas fala sem alma.

- 1 Sem emoção.** Entrega texto falado, não conversa de verdade.
- 2 Um sotaque só.** Nada de carioca, baiano, mineiro, gaúcho. A diversidade do Brasil não existe nas vozes de IA.
- 3 Sem jeitinho.** Falta o ritmo, a gíria e o calor do jeito brasileiro de falar.

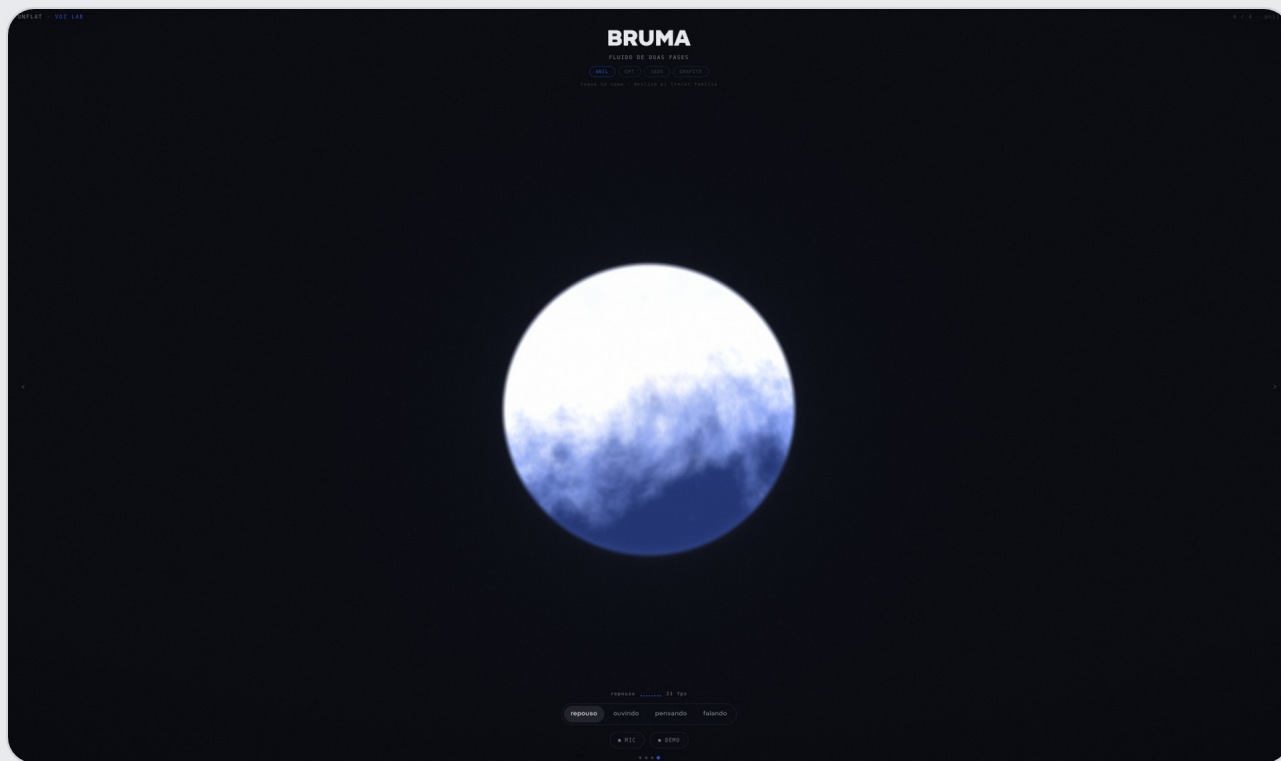
E as vozes com emoção de verdade (Sesame, GPT, Google, ElevenLabs) são estrangeiras, fechadas e caras. Um agente sai cerca de R\$ 0,50 a 0,70 por minuto, tudo em dólar.

A indústria toda é de fora.

	EMOÇÃO AO VIVO	SOTAQUE REGIONAL	BAIXO CUSTO	FOCO EM PT-BR
Sesame	✓	×	×	×
OpenAI · GPT	×	×	×	×
Google	×	✓	×	×
ElevenLabs	✓	✓	×	×
A nossa voz	✓	✓	✓	✓

Nenhuma junta tudo: emoção ao vivo, sotaques regionais brasileiros, custo baixo e foco em pt-BR. O espaço está vazio.

Um agente de voz brasileiro.



BRUMA é a voz e o rosto do nosso agente: um fluido que respira e muda de estado enquanto ouve, pensa e fala.

Conversa com emoção

responde no clima, não lê texto de forma mecânica

Sotaques regionais

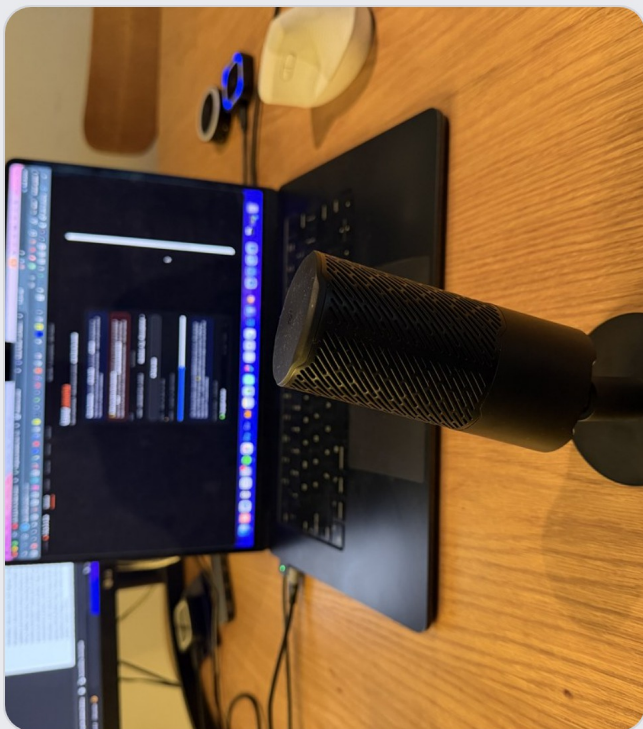
a primeira família de vozes com sotaques do Brasil

Responde na hora

menos de 1 segundo, rápido pra conversa natural

Já estamos gravando o dataset.

Conversas reais, gravadas em estúdio, com três vozes de sotaques diferentes. Cada hora vira áudio limpo, classificado por emoção e trejeitos brasileiros, pronto pro treino.



Pedro

CARIOCA · RJ



João

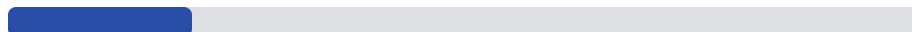
NITEROIENSE · RJ



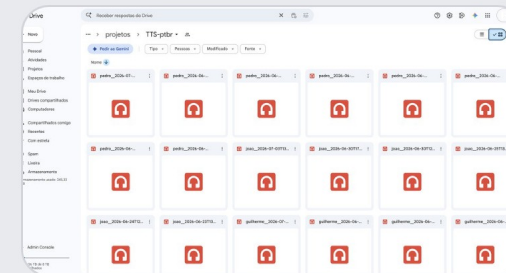
Guilherme

PAULISTANO · SP

30 / 150 horas de áudio limpo



3 vozes, 3 sotaques. gravando toda semana.



A pesquisa por trás da nossa voz.

28 Jul 2024

Moshi: a speech-text foundation model for real-time dialogue

Alexandre Défossez¹
Laurent Mazaud¹
Mam Ousini
Amélie Royer
Patrick Pérez
Hervé Jégou
Edouard Grave^{*}
Neil Zeghidour^{*}
Equal contribution

ALEX@KYUTAIL.ORG
NEIL@KYUTAIL.ORG

Abstract
We introduce Moshi, a speech-text foundation model and full-duplex spoken dialogue framework. Current systems for spoken dialogue rely on pipelines of independent components, namely voice activity detection, speech recognition, textual dialogue and text-to-speech.

22 Jun 2026

Qwen3-TTS Technical Report

Qwen Team

<https://huggingface.co/collections/qwen/qwen3-tts>
<https://modelscope.cn/collections/qwen/qwen3-tts>
<https://github.com/QwenLM/Qwen3-TTS>

Abstract
In this report, we present the Qwen3-TTS series, a family of advanced multilingual, controllable, robust, and streaming text-to-speech models. Qwen3-TTS supports state-of-the-art 3-second voice cloning and description-based control, allowing both the creation of entirely novel voices and fine-grained manipulation over the output speech. Trained on over 5 million hours of speech data spanning 10 languages, Qwen3-TTS adopts a dual-track LLM architecture for real-time synthesis, coupled with two speech tokenizers: 1) Qwen3-TTS-Tokenizer-25Hz is a single-codebook codec emphasizing semantic content, which offers seamlessly integration with Qwen-Audio and enables streaming waveform reconstruction via a block-wise DIT 2; Qwen3-TTS-Tokenizer-12Hz achieves extreme bitrate reduction and ultra-low-latency streaming, enabling immediate first-packet emission (97 ms) through its 12.5 Hz, 16-layer multi-codebook design and a lightweight causal ConvNet. Extensive experiments indicate state-of-the-art performance across diverse objective and subjective benchmarks (e.g., TTS multilingual test set, InstructTTSEval, and our long speech test set). To facilitate community research and development, we release both tokenizers and models under the Apache 2.0 license.

27 Aug 2025

Step-Audio 2 Technical Report

StepFun Audio Team

Abstract
This paper presents Step-Audio 2, an end-to-end multi-modal large language model designed for industry-strength audio understanding and speech conversation. By integrating a latent audio encoder and reasoning-centric reinforcement learning (RL), Step-Audio 2 achieves promising performance in automatic speech recognition (ASR) and audio understanding. To facilitate genuine end-to-end speech conversation, Step-Audio 2 incorporates the generation of discrete audio tokens into language modeling, significantly enhancing its responsiveness to paralinguistic information such as speaking styles and emotions. To effectively leverage the rich textual and acoustic knowledge in real-world data, Step-Audio 2 integrates retrieval-augmented generation (RAG) and is able to call external tools such as web search to mitigate hallucination and audio search to switch timbres. Trained on millions of hours of speech and audio data, Step-Audio 2 delivers intelligence and expressiveness across

20 Jun 2026

Streaming T5-based Text-to-Speech Synthesis with Limited Lookahead

Muyang Du¹, Jason Roche², Junjie Lai¹

¹NVIDIA, China
²NVIDIA, USA
[myoungdu, jroche, julienl}@nvidia.com](mailto:{myoungdu, jroche, julienl}@nvidia.com)

Abstract
Streaming text-to-speech synthesis in cascaded LLM-TTS systems still faces latency challenges as most TTS models require full context before initiating generation. We present S5-TTS, a Streaming variant of T5-TTS that enables low-latency, word-by-word incremental speech synthesis through encoder-decoder language modeling and monotonic alignment learning. S5-TTS begins generating speech immediately after receiving the first few words, substantially reducing end-to-end response latency. To maintain quality under limited lookahead, we introduce a lookahead-causal masking mechanism with Core-based auxiliary attention that preserves intelligibility and speaker similarity, and employ interleaved multi-source distillation to further sentence becomes available. Neural TTS [20] demonstrates the feasibility of neural incremental synthesis by adapting Tacotron [21] with Griffin-Lim [22] and shows that lookahead acoustic context is crucial for natural prosody. Inspired by the prefix-prefill framework for simultaneous translation [23], subsequent work [24] introduces the lookahead policy into Tacotron 2 [25] with Parallel WaveGAN [26], showing that even a single token lookahead substantially improves naturalness. Further analyses [27] quantify this effect, reporting that one-word lookahead already recovers 80% of the full-context representation, while two-word lookahead increases this to 94%. Speech-T2 [28] extends the idea with a transducer-based architecture for unified ASR and TTS, likewise confirming that lookahead is sufficient to close the gap with non-streaming quality. More re-

Jul 2026

Transformer-based segmentation of prosodic boundaries in Brazilian Portuguese

¹ Rodrigo de Freitas Lima
ICMC
University of São Paulo
São Paulo, Brazil
<https://orcid.org/0009-0009-4344-1109>

² Julio Cesar Galdino
ICMC
University of São Paulo
São Paulo, Brazil
<https://orcid.org/0000-0001-6378-4648>

³ Marcos Vinícius Trevisio
ICMC
Instituto Superior Técnico
University of Lisbon
Lisbon, Portugal
<https://orcid.org/0000-0002-3286-0609>

Abstract
Automatic prosodic segmentation identifies boundaries between speech units from acoustic and linguistic evidence. Although recent deep learning approaches have produced strong results for English, automatic segmentation for Brazilian Portuguese (BP) still relies mostly on rule-based or traditional machine-learning methods. This paper presents SAMPA, a Whisper-based segmenter that transcribes BP speech while inserting explicit markers for terminal prosodic boundaries. We fine-tune Whisper-large-v3 on manually segmented recordings from the NUSRC-SP dataset and evaluate different training and test-time filtering configurations, including out-of-distribution testing on the MarE-Diversidades dataset. SAMPA achieves competitive boundary-detection performance across settings, with the best models reaching $F_1 = 0.71$ on the held-out test split and

6 Nov 2025

The Impact of Prosodic Segmentation on Speech Synthesis of Spontaneous Speech

Julio Galdino¹0000-0001-6378-4648, Sidney Leal²0000-0002-0017-2003, Letícia de Souza³0000-0003-7310-0290, Rodrigo Lima¹0000-0001-4344-1109, Antonio Moreira¹0000-0001-9007-2100, Arnaldo Cundido Jr.¹0000-0002-3402-0003, Miguel Oliveira Jr.¹0000-0002-0001-0003, Edresson Casanova¹0000-0003-0008-7570, and Sandra Ajudiz¹0000-0001-1100-0001

¹ University of São Paulo, São Carlos, SP, Brazil
² Universidade Estadual Paulista, São João do Rio Preto, SP, Brazil
³ Universidade Federal de Alagoas, Maceió, AL, Brazil
⁴ NVIDIA Corporation, São Paulo, SP, Brazil
⁵ Venturus - Centro de Inovação Tecnológica, Campinas, SP, Brazil
Corresponding author: julio@lml.inf.br, sidley@gmail.com

Abstract. Spontaneous speech presents several challenges for speech synthesis, particularly in capturing the natural flow of conversation, including turn-taking, pauses, and disfluencies. Although speech synthesis

16 Mar 2026

TAGARELA - A PORTUGUESE SPEECH DATASET FROM PODCASTS

Frederico Santos de Oliveira¹
Augusto Sehn da Rosa¹
Christopher Dane Shulby¹
Anderson da Silva Soares¹

Lucas Rafael Sefkanal Gris¹
Alexandre Costa Ferro Filho¹
Rafael Teixeira Sousa²
Arifindo Rodrigues Galvão Filho¹

Alief Iury Siqueira Ferreira¹
Edresson Casanova¹
Diogo Fernandes Costa Silva¹

¹ Federal University of Mato Grosso (UFMT) ² Federal University of Goiás (UFG)
³ Paulista State University (UNESP) ⁴ NVIDIA ⁵ Elsa Speak

ABSTRACT
Despite significant advances in speech processing, Portuguese remains under-researched due to the scarcity of public, large-scale, and high-quality datasets. To address this gap, we present a new dataset, named TAGARELA, composed of over 8,972 hours of podcast audio, specifically curated for training automatic speech recognition (ASR) and text-to-speech (TTS) models. Notably, its scale rivals English's Gigaspeech (100k), enabling state-of-the-art Portuguese models. To ensure data quality, the corpus was subjected to an audio re-

30 Apr 2023

YourTTS: Towards Zero-Shot Multi-Speaker TTS and Zero-Shot Voice Conversion for everyone

Edresson Casanova¹, Julian Weber², Christopher Shulby³, Arnaldo Candido Junior⁴, Eren Gölgör⁵ and Mouez Antonelli Pons¹

¹ Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, Brazil
² Sopra Banking Software, France
³ Defined.ai, United States of America
⁴ Federal University of Technology - Paraná, Brazil
⁵ Coqui, Germany
edresson@usp.br

Abstract
YourTTS brings the power of a multilingual approach to the task of zero-shot multi-speaker TTS. Our method builds upon the VITS model and adds several novel modifications for zero-shot multi-speaker and multilingual training. We achieved state-of-the-art (SOTA) results in zero-shot multi-speaker TTS and results comparable to SOTA in zero-shot voice conversion on the VCTK dataset. Additionally, our approach achieves promising results in a tandem task with a single-speaker dataset, even

No que o Centelha vira.

R\$ 128k

para a primeira voz de IA com sotaque brasileiro.

+ R\$ 45,6k bolsa CNPq (equipe)
12 meses de execução

■ **Treinamento de IA, 100% local**

GPU própria pra treinar os modelos aqui, sem nuvem. A nuvem vem depois, só pro SaaS.

■ **Dataset + desenvolvimento**

captar e classificar o máximo de áudio em alta qualidade, com sotaques diversos. 150h a gente já faz sozinho, mirando 1500h.

■ **Reserva + protótipos**

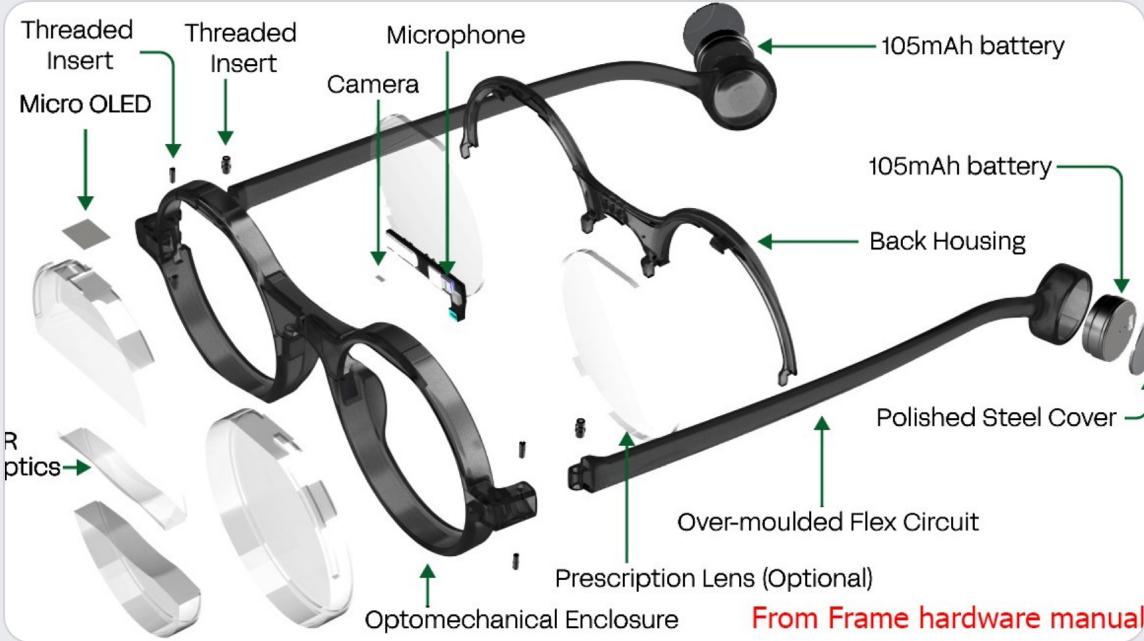
reserva e prototipagem com hardwares de voz.

Onde essa voz entra.

- **Acessibilidade e inclusão**
quem tem deficiência visual ou não lê texto
- **Atendimento ao cidadão**
serviços públicos que falam com o povo
- **SAC, varejo e economia criativa**
atendimento e conteúdo em escala
- **Educação**
aulas e audiolivros com voz humana
- **Óculos inteligentes e dispositivos**
guias de áudio e novas interfaces

SOBERANIA · CULTURA · ACESSO

Uma infraestrutura nacional de voz, que não manda dados nem dinheiro pra fora.



ÓCULOS INTELIGENTES

Brilliant Labs Frame · vista explodida (imagem: kgutttag.com)